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Paper

Image Annotation with Incomplete Labelling by Modelling Image Specific Structured Loss

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In this paper, we address the problem of image annotation with *incomplete labeling*, where multiple objects in each training image are not fully labeled. The conventional one-versus-all support vector machine (OVA-SVM), which performs fairly well on full labeling, decays drastically under the setting of incompleteness. Recently, a structured output learning method termed OVA-SSVM was proposed to boost the performance of OVA-SVM by modeling the structured associations of labels and show efficiency under the setting of incompleteness. OVA-SSVM assumes that each training sample includes a single label and adopts an loss measure of classification style where, as long as one of the predicted label is correct, the overall prediction should be considered correct. However, this may not be appropriate for the multilabel annotation task. Therefore, we extend the OVA-SSVM method to the multilabel situation and design a novel image-specific structured loss to account for the dependences between predicted labels relying on image label associations. The superiority of the proposed image-specific structured loss is that it can directly learn the semantic relationships of labels from training data without predefined semantic hierarchy. Extensive empirical results on a variety of benchmark datasets show that the proposed method performs significantly better than OVA-SSVM on image annotation tasks with incomplete labeling and achieves competitive performance compared to other state-of-the-art methods. © 2015 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

Keywords: image annotation, structured output learning, incomplete labeling

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1. Introduction and literature review

Recently, with the ever-growing popularity of social media, the number of images with user-provided labels available over the Internet has increased dramatically. For example, Flickr has more than two billion images with millions of newly uploaded photos per day. One approach to retrieve or manage such a large number of images is to automatically annotate each test image with multiple labels by training an annotation model on a manually labeled training set. This problem is termed *image annotation*, which has emerged as an important research area during the last decade, and numerous attempts have been made to improve its accuracy.

Generally, image annotation can be viewed as a multilabel classification problem, as each image is associated with multiple labels. Several approaches cast the problem of multilabel classification to separately train a discriminative classifier for each label, and then use these classifiers to predict the presence/absence of each particular label. Usually, support vector machine (SVM) [1], boosting [2], and random forest [3] are typically applied to learn the discriminative classifiers. It is worth noting that the one-versus-all SVM (OVA-SVM), which has been widely used on image classification tasks [4,5], also shows promising results on image annotation task, and hence it is usually treated as a baseline for annotation model evaluation. However, these approaches neglect the correlation of different labels that are useful for model learning. To exploit the dependence among labels, several efficient schemes have been proposed in the multilabel classification framework: e.g. incorporating the correlation of labels from prior label information [6]; using hypergraph regularization to constrain the label correlation [7]; and constructing bi-relational graph to model the label correlation and image similarity [8]. The main shortcoming of multilabel classification methods is that, in order to train a reliable annotation model, they require a large set of training images with clean and complete manual annotations.

Besides the multilabel classification methods, several advanced machine learning techniques have been applied, including generative methods and nearest neighbor (NN)-based methods. The generative methods [9–11], such as topic model [11], learn the joint distribution of labels and image features, where different hypotheses of probability distribution can be assumed, such as multinomials, separate Bernoullis, or mixture of Gaussian. The NN-based methods [12–15] perform image annotation via propagating labels from visually similar images, since such methods generally assume that visually similar images probably share common labels. For the NN-based methods, the primary research issue focuses on designing efficient label propagation schemes [12,13,16] and selecting appropriate visual neighbors [14,15,17]. Similar to the multilabel classification methods, to achieve good performance, the generative and NN-based methods also require a large number of well-annotated images.

From the three types of methods discussed above, it can be learnt that the quality of human annotations of the training set is quite crucial in training a good annotation model as they provide informative knowledge of image–label associations. However, one cannot expect to obtain a complete set of labels for a training

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(a) Groundtruth: bloom, leave (b) Groundtruth: clouds

Fig. 1. Examples of incomplete labeling: two images are from benchmark datasets IAPRTC-12 [20] (a) and NUS-WIDE [21] (b). Potentially correct labels such as {*flower, plant, tree, trunk*}, {*sky, grass*} are missed from the groundtruth of two images, respectively

image, especially for images extracted from social media [18,19], since human labelers usually annotate only prominent labels and typically miss out several objects present in the image. Here we pose a practical issue of *incomplete labeling*, where the images in the training set are not completely annotated with all relevant labels. As shown in Fig. 1, the two images from benchmark datasets (collected from social media such as Flickr) have few annotated labels and suffer from the problem of incomplete labeling.

Thus, when applying these methods on incompletely labeled dataset, the annotation performance can hardly achieve optimal results, since the insufficient annotations of the dataset could potentially lead to a biased estimation of the prediction model. For example, a recent study [22] reports that the annotation performance of the baseline method OVA-SVM decays drastically when the degree of incompleteness of training set increases.

In this work, we are motivated to develop an annotation method that is more efficient and robust with an incompletely labeled dataset. To tackle the issue of incomplete labeling, one group of recent ongoing researches [23–26] directly modify conventional image annotation prototypes such as multilabel ranking [23,26], binary SVM [24], and ridge regression [25] by incorporating additional consistency between visual and semantic cues in images and labels. And the performance of these methods largely depends on the assumption of consistency. Moreover, another group of works aims at boosting the conventional annotation models and adding a new learning stage under the setting of incompleteness incrementally. The techniques utilized in the new learning stage could be multitask learning [27], ensemble learning [28], and structured output learning [29,30].

In particular, the structured output learning technique has been attracting much attention since it allows the learning of functions with complex outputs, such as object poses, segmentations, and parse trees. And it has been successfully applied in problems such as object detection, image segmentation, and image parsing. Here we would like to emphasize that the structured output learning technique is an efficient scheme to handle the difficulties of incomplete labeling in the image annotation task. First, it captures the interdependencies of multiple labels from the structure in the output space. Second, the weak learning manner allows it to explore the potential usages of missing labels, and those missing labels can be captured by latent variables [31].

Specifically, in the celebrated work termed OVA-SSVM [30], a structured output learning method is adopted to boost the performance of pretrained OVA-SVM classifiers under the setting of incompleteness, and it designs a structured loss function of the image classification style to benefit the prediction of missing labels. And promising prediction results are obtained on the ImageNet dataset [32], where each training image has a single label. However, the OVA-SSVM method may not be well extended to more practical circumstances where each training image has multiple labels, because of the limitation of the structured loss function used in it.

Therefore, in this paper, we attempt to improve OVA-SSVM in three ways: (i) We propose a method that can tackle a more general case where each training image has one or more labels. The proposed method is more efficient for the multilabel annotation problem than the OVA-SSVM method. (ii) We use a novel image-specific structured loss term that is more appropriate than the classification-style structured loss used in OVA-SSVM, to account for the dependencies of predicted multiple labels of a specific image. (iii) We develop an efficient optimization algorithm with lower complexity by exploiting the properties of the proposed structured loss. Finally, we conduct experiments to compare the proposed method with OVA-SSVM on image classification task, observing that the proposed method performs on par with OVA-SSVM and can automatically capture the semantic hierarchies without predefinition (used in OVA-SSVM). We further make extensive evaluation on two benchmark annotation datasets with a variety of the setting of incompleteness for the image annotation task. The empirical results demonstrate that the proposed method is significantly better than OVA-SSVM and achieves competitive annotation performance compared to other state-of-the-art methods designed for incomplete labeling.

2. Problem formulation

In this section, we will first introduce the conventional OVA-SVM used for image annotation task, and then describe the OVA-SSVM method that uses structured output learning for boosting the conventional OVA-SVM classifiers under the setting of incompleteness. Some notations used in the following sections are also defined in this section.

2.1. Conventional OVA-SVM

We are given an incompletely labeled dataset $\mathcal{T} = \{(x^1, Y^1), ..., (x^N, Y^N)\}$. Here, $x^n \in \mathcal{X}$ represents the image feature vector, and $Y^n \subseteq \mathcal{Y}$ is a set of labels, where $\mathcal{Y} = \{y_1, ..., y_C\}$ is the vocabulary of C labels. Note that Y^n is a subset of the ideally full set Ω^n of groundtruth labels for image x^n. Our goal is to learn an annotation model that, for an unseen image x, outputs an optimal set $\hat{Y}$ that includes K distinct labels. Usually, K is set to be a fixed value for different test images to simplify the problem setting and model evaluation.

A conventional annotation model consists of learning a series of binary OVA-SVM classifiers that distinguish a single label from all others. For a given label y_c, we denote the parameter vector of learnt OVA-SVM classifier as $\boldsymbol{w}_{OVA}^{y_c}$. Figure 2(a) shows the learning procedure of OVA-SVM and the learnt binary classifiers $\{w_{OVA}^{y_c}\}_{c=1}^{C}$. To predict a set of K labels $\hat{Y}$ for an unseen image x, the annotation model simply returns the labels with the K highest scores performing on classifiers of all labels:

$$\hat{Y} = \arg\max_{Y \subseteq \mathcal{Y}} \sum_{y_c \in Y} x \cdot \boldsymbol{w}_{OVA}^{y_c} \tag{1}$$

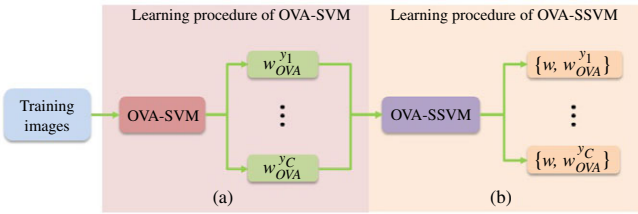


Fig. 2. Procedure of parameter learning for OVA-SVM (a) and OVA-SSVM (b) methods

where $Y \subseteq \mathcal{Y}$ represents any output set containing K labels. It is worth noting that the annotation model of OVA-SVM classifiers is suboptimal since (i) the one-versus-all learning manner ignores the dependencies of labels, which implies that the OVA-SVM optimizes the prediction of only one single output label, ignoring the "structure" altogether, and (ii) the performance of OVA-SVM classifiers would drop drastically when incomplete labels for training the image are provided.

2.2. OVA-SSVM To overcome the disadvantages of conventional OVA-SVM and to exploit the structured associations in output label set Y, the structured output learning method OVA-SSVM [30] considers that the training set consists of structured input–output pairs $\mathcal{T} \in (\mathcal{X} \times \mathcal{Y})^N$. The prediction rule of optimal output labels $\hat{Y}$ for an unseen image x is

$$\hat{Y} = \arg\max_{Y \subseteq \mathcal{Y}} \Phi(x, Y) \cdot \boldsymbol{w} = \arg\max_{Y \in \mathcal{Y}} \sum_{y \in Y} \phi(x, y) \cdot \boldsymbol{w} \tag{2}$$

where Φ is the joint feature vector that describes the relationship between input x and any structured output Y, ϕ is the joint feature vector for input x and a single label y in Y, and $\boldsymbol{w}$ is the parameter vector to be learnt. In particular, given a set of pretrained OVA-SVM classifiers $\{\boldsymbol{w}_{OVA}^{y_c}\}_{c=1}^{C}$, the joint feature vector $\Phi(x, Y)$ in OVA-SSVM is defined as

$$\Phi(x, Y) = \sum_{y \in Y} x \circ \boldsymbol{w}_{OVA}^{y} \tag{3}$$

where $x \circ \boldsymbol{w}_{OVA}^{y}$ represents the Hadamard product of x and $\boldsymbol{w}_{OVA}^{y}$. Then the annotation model in (2) can be formulated as

$$\hat{Y} = arg \max_{Y \subseteq \mathcal{Y}} \sum_{y \in Y} \langle x \circ \boldsymbol{w}_{OVA}^{y}, \boldsymbol{w} \rangle \tag{4}$$

We can learn from (4) that OVA-SSVM incrementally learns a single parameter vector $\boldsymbol{w}$ that re-weights the parameter vectors of existing OVA-SVM classifiers $\{\boldsymbol{w}_{OVA}^{y_c}\}^{y_c \in \mathcal{Y}}$ (the learning procedure of OVA-SSVM is illustrated in Fig. 2(b), and incorporates the structured nature of output Y through the joint feature vector $\Phi(x, Y)$.

Moreover, for the setting of incomplete labeling of $\mathcal{T}$, the input-output relationship is not completely characterized by $(x, Y) \in \mathcal{X} \times \mathcal{Y}$. It is rational to introduce a set of unobserved latent variables $Z = \{Z^1, ..., Z^N\}$, where Z^n represents the set of labels that appear in the image x^n but were not annotated in the groundtruth. The full set of labels for the image x^n is $\Omega^n = Y^n \cup Z^n$ (note that $Y^n \cap Z^n = \varnothing$). Now the joint feature vector $\Phi(x, \Omega)$ describes the relation among input x, output Y, and latent variables Z, and it is defined as

$$\Phi(x, \Omega) = \sum_{y \in Y} x \circ \boldsymbol{w}_{OVA}^{y} + \sum_{z \in Z} x \circ \boldsymbol{w}_{OVA}^{z} \tag{5}$$

To train OVA-SSVM, the parameter vector $\boldsymbol{w}$ is determined by minimizing the regularized risk on the training set $\mathcal{T}$. Risk is measured through a user-provided structured loss function $\Delta(Y, Y^n)$, which quantifies how much the prediction Y differs from the given label set Y^n of the image x^n. The resulting convex optimization problem is to minimize the objective function as

$$\min_{\boldsymbol{w}, \xi} \frac{\lambda}{2} \|\boldsymbol{w}\|^2 + \frac{1}{N} \sum_{n=1}^{N} \xi_n \tag{6}$$

$$s.t \quad \boldsymbol{w} \cdot \Phi(x^n, \Omega^n) - \boldsymbol{w} \cdot \Phi(x^n, Y) \geq \Delta(Y, Y^n) - \xi_n, \quad \forall n, Y \subseteq \mathcal{Y}.$$

The constraints of (6) identify the prediction Y with a score $\boldsymbol{w} \cdot \Phi(x^n, Y)$ that is smaller than the score $\boldsymbol{w} \cdot \Phi(x^n, \Omega^n)$ of the "full" groundtruth Ω^n by a soft margin equal to the loss $\Delta(Y, Y^n)$ with the slack variable ξ_n. The optimization problem can be solved efficiently using a constraint generation strategy [33]: we can generate the constraint by identifying the most violated (incorrect) prediction $\overline{Y}$ from Y for the current parameter vector $\boldsymbol{w}$ on x^n. This amounts to solving

$$\overline{Y} = \arg\max_{Y \subseteq \mathcal{Y}} \{\Delta(Y, Y^n) + \boldsymbol{w} \cdot \Phi(x^n, Y)\} \tag{7}$$

Given the definition of user-provided structured loss Δ, we can use $\overline{Y}$ of all $x^n \in \mathcal{X}$ to approximate a lower bound of the objective in (6). Then we can compute the gradient of (6), or, alternately, optimize the latent variables Z and the parameter vector $\boldsymbol{w}$. In the next section, we will introduce the proposed image-specific loss term that is elaborately designed for incompletely labeled training data, and derive the corresponding optimization algorithm in the structured output learning framework.

3. Proposed structured loss under the setting of incompleteness

3.1. Image specific structured loss Since the given label set Y^n may not describe all the object in image x^n, an annotation model should not be penalized for predicting "incorrect" labels that actually describe those objects in x^n. To address this issue, a structured loss function Δ is designed in OVA-SSVM. It argues that, given a set of predicted output labels Y for x^n, it should not penalize the method if one of the predicted labels $y \in Y$ is similar to any of the groundtruth labels $y^n \in Y^n$. The loss function is defined as

$$\Delta(Y, Y^n) = \min_{y \in Y} \min_{y^n \in Y^n} d(y, y^n) \tag{8}$$

where $d(y, y^n)$ is the error term measuring the difference between label y and y^n. In practice, $d(y, y^n)$ could be a flat error measure: $d(y, y^n) = 0$ if $y = y^n$, and 1 otherwise. Besides, $d(y, y^n)$ could also be a hierarchical error measure that measures the shortest distance between y^n and y in a taxonomic vocabulary tree.

Actually, there are several limitations of the structured loss of (8) for the setting of incompleteness. First, to predict output labels Y, ensuring that only one of the predicted labels is similar to the groundtruth is not enough. In other words, it is expected that each of the predicted labels is similar to any (even all) of the groundtruth labels. Second, the error measure of $d(y, y^n)$ is either too coarse to quantify the difference of labels (i.e. flat error measure) or too rigorous to require the prior construction of taxonomic tree (i.e. hierarchical error measure). Third, the error measure indicates that the variances of labels are based on the global statistics of training data, whereas for the setting of incompleteness it is not sufficient to model the relatedness of missing labels and groundtruth labels.

In Fig. 3, we demonstrate two examples of label prediction using *flat* structured loss in OVA-SSVM. It can be observed that, although flat structured loss (in fifth column) is generated to be zero (since the predicted labels "bloom", "man" match the incomplete groundtruth Y^n), the predicted result (in third column) is inferior since it contains several incorrect labels, e.g. {"fruit", "forest"}, {"woman", "bottle", "forest"}. Thus, it implies that numerically minimizing the structured loss of (8) could not guarantee all predicted labels to be similar to groundtruth labels.

To address the limitations of (8), we assume that each of the predicted labels is related to *all* of the groundtruth labels, and we desire the structured loss term to capture the variances of labels relying on the specific image content. Our proposed image-specific loss function is formulated as

$$\Delta(Y, Y^n; x^n) = \frac{1}{|Y|} \frac{1}{|Y^n|} \sum_{y \in Y} \sum_{y^n \in Y^n} d(y, y^n; x^n) \tag{9}$$

Image x^n	Incomplete lables Y^n	Predicted labels Y		Structured loss Δ	
		OVA-SSVM (Flat)	Proposed	Flat	Image specific
	bloom, leave	bloom, flower, fruit, forest, branch	bloom (0), leave (0), trunk (0.531), flower (0.552), plant (0.765)	0	0.3696
	man, one, rock	man, woman, front, bottle, forest	man (0), rock (0), tee-shirt (0.647), hand (0.685), waterfall (0.689)	0	0.4042

Fig. 3. Examples of label prediction using flat loss (OVA-SSVM (Flat)) vs. image-specific structured loss (proposed method). These two images are selected from the IAPRTC-12 dataset. Note that in the fourth column, the image-specific loss of each predicted label is also presented. And the loss values Δ in last two columns are calculated according to (8) and (9) respectively

Here, the error measure $d(y, y^n; x^n)$ is image-specific, representing the difference of label y and y^n particularly on image x^n. In addition, the structured loss $\Delta(Y, Y^n; x^n)$ ensures that each of the predicted labels in Y is related to all the groundtruth labels in Y^n. Since the incomplete label set Y^n is small, here we restrict the structured loss of (9) to moderately consider the dependencies between each of the predicted labels and all labels in Y^n.

Inspired by the works in Refs [24,34], we cast measuring $d(y, y^n; x^n)$ to comparing the relatedness of image x^n to labels y and y^n. In particular, for a given label y_c, let $\mathcal{X}_c^+$ be the set of images that are annotated with label y_c, and the remaining images as $\mathcal{X}_c^- = \mathcal{X} \backslash \mathcal{X}_c^+$. For image x^n in $\mathcal{X}_c^+$, we define the relatedness of image x^n to label y_c as $R(x^n, y_c) = 1$ since x^n is annotated with y_c. And for image x^n belonging to $\mathcal{X}_c^-$ of y_c, we determine the relatedness score of $R(x^n, y_c)$ considering three factors: visual similarity, semantic similarity, and image–label association in the visual neighborhood. Correspondingly, $R(x^n, y_c)$ consists of

- Visual similarity-based relatedness score $R_V(x^n, y_c)$: We compute the visual distance $dist(\cdot)$ (scaled to range $[0, 1]$) of x^n with its nearest neighbor $x^* \in \mathcal{X}_c^+$, and define $R_V(x^n, y_c) = 1 - dist(x^n, x^*)$.
- Semantic similarity-based relatedness score $R_S(x^n, y_c)$: We first compute the correlation score between pairwise labels y_i and y_j, $\forall y_i, y_j \in \mathcal{Y}$ as: $co_occur(y_i, y_j) = \frac{f_{i,j}}{f_i + f_j - f_{i,j}}$, where f_i and f_j are the count of occurrence of labels y_i and y_j, and $f_{i,j}$ is the count of co-occurrence of labels y_i and y_j. Let Y^n be the label set of image x^n, and we define $R_S(x^n, y_c) = \max_{y^n \in Y^n} co_occur(y_c, y^n)$.
- Reverse nearest neighbor-based relatedness score $R_N(x^n, y_c)$: Given a fixed value of $M (= 5)$, we first calculate the mth nearest neighbor for each image in $\mathcal{X}_c^+$ on the whole training set $\mathcal{X}$, and then we set p_m to be the number of images in $\mathcal{X}_c^+$ that have x^n as their mth nearest neighbor. Finally, we define $R_N(x^n, y_c) = \sum_{m=1}^{M} \frac{p_m}{m} / \sum_{m=1}^{M} p_m + \varepsilon$, where $\varepsilon > 0$ is a small number to avoid division by zero.

Finally, $R(x^n, y_c)$ is defined as the average of these three scores, as in [24]:

$$R(x^n, y_c) = \frac{1}{3}(R_V(x^n, y_c) + R_S(x^n, y_c) + R_N(x^n, y_c)) \tag{10}$$

Now we can calculate the error measure $d(y, y^n; x^n)$ by comparing the relatedness scores of image x^n to labels y and y^n as

$$d(y, y^n; x^n) = R(x^n, y^n) - R(x^n, y) = 1 - R(x^n, y) \tag{11}$$

Recalling that $y^n \in Y^n$ is the groundtruth label of x^n, it has the highest relatedness score (equal to 1). It can be seen that the calculation of (11) is directly determined by the relatedness score $R(x^n, y)$ of label y to image x^n. And if the predicted label y has a larger relatedness score to x^n, it would have a smaller difference with all the groundtruth labels. This is consistent with the proposed structured loss of (9), which now can be efficiently measured by the relatedness of predicted labels to the specific image.

Compared to the flat/hierarchical structured loss, our proposed structured loss of (9) has several advantages. First, as shown in Fig. 3, although the loss values (in the last column) are numerically larger than "zero" of flat structured loss (in the fifth column), the predicted labels are more similar to the provided incomplete labels. This is because the proposed structured loss moderately considers the predicting labels based on their relatedness to specific image content, and the relatedness measure is elaborately designed and more appropriate than the simple 0–1 measure. Second, the proposed structured loss is more flexible to the number of groundtruth labels, as it accumulatively measures each of the predicted labels to all the groundtruth, while the flat structured loss focuses on the most dominant one in the predicted label to a single label of the groundtruth. Third, the relatedness measure can be directly and precisely computed from labeled training images, while prior knowledge of taxonomy or large quantities of training data with full labeling is required in the traditional algorithms.

3.2. Optimization method Given the proposed structured loss function of (9), we can generate the most violated constraint of prediction $\overline{Y}$ for image x^n according to (7) in the form

$$\begin{aligned}\overline{Y} &= \arg\max_{Y \in \mathcal{Y}} \{\frac{1}{|Y|}\frac{1}{|Y^n|}\sum_{y \in Y}\sum_{y^n \in Y^n} d(y, y^n; x^n) + \sum_{y \in Y} \boldsymbol{w} \cdot \phi(x^n, y)\} \\ &= \arg\max_{Y \in \mathcal{Y}} \{\frac{1}{|Y|}\sum_{y \in Y}(1 - R(x^n, y)) + \sum_{y \in Y} \boldsymbol{w} \cdot \phi(x^n, y)\},\end{aligned} \tag{12}$$

where the calculation of structured loss $\Delta(Y, Y^n; x^n)$ is converted to computing the relatedness scores of predicted label set Y to image x^n, as described in Section 3.1.

We can obtain the solution of Y of (12) by simply sorting the term $\frac{1}{|Y|}(1 - R(x^n, y_c)) + \boldsymbol{w} \cdot \phi(x^n, y_c)$ for each label $y_c \in \mathcal{Y}$, and then choosing the top K labels for $\overline{Y}$. Solving (12) greedily takes $\mathcal{O}(C \log C)$, thus it is faster than the constraints generation method in OVA-SSVM, which takes $\mathcal{O}(C^2 \log C)$. After generating the most violated constraint $\overline{Y}$ for each image, the lower bound of the objective function in (6) can be derived as

$$\begin{aligned}J(\boldsymbol{w}) = \frac{\lambda}{2}\|\boldsymbol{w}\|^2 + \frac{1}{N}\sum_{n=1}^{N}[\Delta(\overline{Y} - Y^n) \\ + \boldsymbol{w} \cdot \Phi(x^n, \overline{Y}) - \boldsymbol{w} \cdot \Phi(x^n, \Omega^n)]\end{aligned} \tag{13}$$

and the gradient of $J(\boldsymbol{w})$ with respect to $\boldsymbol{w}$ is

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \lambda \boldsymbol{w} + \frac{1}{N}\sum_{n=1}^{N}\left[\boldsymbol{w} \cdot \Phi(x^n, \overline{Y}) - \boldsymbol{w} \cdot \Phi(x^n, \Omega^n)\right] \tag{14}$$

It can be observed in (13) and (14) that calculating $J(\boldsymbol{w})$ and its gradient involves in computing the joint feature vector $\Phi(x^n, \Omega^n)$ on 'full' label set Ω^n of each image. Here the number of labels in Ω^n is fixed as K so that the missing labels Z^n can be precisely inferred. Thus, $\Phi(x^n, \Omega^n)$ can be efficiently computed according to (5) with the latent variable Z^n. To learn the parameter vector $\boldsymbol{w}$ with latent variable Z^n, we follow the previous alternating optimization technique proposed in [29,30]. Specifically, we alternate between optimizing the parameter vector

Input: Incompletely labeled training data $\mathcal{T} = \{(x^n, Y^n)\}_{n=1}^N$, pre-trained binary classifiers $\{\boldsymbol{w}_{OVA}^{y_c}\}_{c=1}^C$
Output: Parameter vector $\boldsymbol{w}$
Initialize $\boldsymbol{w}_0 = \mathbf{1}$ for iteration $t = 0$
repeat
  Set $t = t + 1$
  for $n = 1, ..., N$ **do**
    Assign latent variable $Z_t^n = \{arg\max_{Y \in \mathcal{Y}} \boldsymbol{w}_{t-1} \cdot \Phi(x^n, Y)\} \backslash Y^n$ for x^n (preserving $K - |Y^n|$ missing labels)
  end for
  for $n = 1, ..., N$ **do**
    Generate the most violated constraint $\bar{Y}_t$ for x^n according to Eq.12
  end for
  Compute objective $J_t(\boldsymbol{w})$ and gradient $\nabla_{\boldsymbol{w}} J_t(\boldsymbol{w})$ according to Eq.13 and Eq.14
  Minimize loss of Eq.6 to calculate $\boldsymbol{w}_t$
until Loss in Eq.6 is converged

Fig. 4. Alternating optimization of the proposed method

$\boldsymbol{w}^t$ by initializing the latent variable Z^n for each image in the tth iteration, and re-estimating the latent variable Z^n for the $(t+1)$th iteration given the learnt parameter vector $\boldsymbol{w}^t$. The pseudocode for solving the alternating optimization problem is depicted in Fig. 4.

4. Experimental results

In this section, we first directly compare the proposed method with previous OVA-SSVM on image classification task and explore the effectiveness of the proposed image-specific structured loss function. Then we further evaluate the proposed method by comparing it with OVA-SSVM and other state-of-the-art annotation methods on image annotation task under a variety of setting of incompleteness.

4.1. Experimental setup

4.1.1. Datasets and Features For image classification task, we adopt the dataset of the ImageNet Large Scale Visual Recognition 2010 Challenge (ILSVRC-10) [32], which has been used in previous OVA-SSVM methods. This dataset contains 1.2M training images of 1000 object classes (with between 660 to 3047 images per class) and a test set of 150K images. Each image in this dataset is labeled with a single positive class. For image annotation task, we use two publicly available benchmark datasets: IAPRTC-12 [20] and NUS-WIDE [21]. These two datasets are very challenging with significant diversity among the images that are obtained from the social web, and they both cover the properties of large vocabulary size and large number of images. Note that each image in these two datasets is annotated with multiple labels. Table I summarizes the general statistics of all the three datasets.

To represent an image, for ILSVRC-10 dataset we choose the similar Bag-of-Words (BoW) model as in OVA-SSVM method: dense SIFT feature are extracted from each image and quantized using a visual vocabulary of 1024 visual words. Next, the visual words are pooled in a histogram, and then normalized with L2 normalization to a 1024-dimensional vector to represent each image. This representation is widely used as the baseline in the image classification methods [4,5]. Furthermore, recently, the deep convolutional neural network (CNN) [35] is broadly adopted to achieve the state-of-the-art performance for large-scale image classification tasks. To leverage the already good performance of the existing method, we use the popular deep learning framework Caffe [36] and the model trained on the ILSVRC-12 dataset to extract CNN features for training/testing images of the ILSVRC-10 dataset. Specifically, we extract the highest level feature of the network ("fc7" layer in Caffe model), which resulting in a 4096-dimensional CNN feature vector for one image.

Table I. General statistics for the three datasets used in experiment. The items in the second row are listed in the format "training/test", and items in the third and fourth rows are given in the format "mean/minimum/maximum"

	ILSVRC-10	IAPRTC-12	NUS-WIDE
Total labels	1000	291	81
No. of images	1,200,000/ 150,000	17,665/1,962	138,563/ 92,484
Labels per image	1/1/1	5.7/1/23	1.8/1/20
Images per label	1,200/660/ 3,047	347/153/4,999	2,512/ 333/16,425

For IAPRTC-12 and NUS-WIDE datasets, we use a combination of multiple types of feature descriptors, since such a scheme has been reported in several state-of-the-art image annotation methods [12,13,24–26] where fusing multiple visual features can improve the annotation performance of using a single visual feature. Specifically, for the IAPRTC-12 dataset, we use the same multiple features as those in [12,13,24,25]. These multiple features consist of global and local features. The global features include histograms in RGB, HSV, and LAB color space, and the GIST [37] features; and the local features include the SIFT and hue descriptors extracted densely from multiscale grid and from Harris–Laplacian interest points. For the NUS-WIDE dataset, besides global GIST features, we also utilize five types of SIFT-based local features (C-SIFT, Opponent-SIFT, RGB-SIFT, RG-SIFT) using the public colorDescriptor tools [38]. For these two datasets, we first separately perform L2 normalization for each type of feature, and then concatenate them to a feature vector (37 152 dimensions for IAPRTC-12 and 20 512 dimensions for NUS-WIDE) to represent each image.

4.1.2. The setting of incompleteness We consider the original IAPRTC-12 as a fully labeled dataset since the average number of labels per image is more than 5 (5.7 in Table I), which is sufficient to describe multiple objects in an image. To simulate the setting of incompleteness, we randomly delete some of the labels for each image, and the deletion process stands by the principle $\min(1, \lceil M \times (1 - ratio) \rceil)$, ensuring that each image preserves at least one label. Here M denotes the number of original labels of an image, $\lceil \cdot \rceil$ denotes the ceiling function that gives the smallest integer not smaller than the given value, and *ratio* represents the degree of incompleteness. In our experiments, we set $ratio = \{10\%, 30\%, 50\%, 70\%, 90\%\}$, indicating that the larger the ratio, the higher the degree of incompleteness. For NUS-WIDE, the average number of labels per image is less than 2 (1.8 in Table I), which could be insufficient compared with the situation of IAPRTC-12. Therefore we treat NUS-WIDE as an incompletely labeled dataset, and directly utilize the original annotations for the setting of incompleteness.

4.1.3. Binary classifiers As the proposed method needs pretrained binary classifier for each class as a starting point for structured output learning, we follow previous works [24,30] and learn OVA-SVM classifiers for the setup. In particular, we train a linear OVA-SVM classifier for each label using the Pegasos [39] algorithm and calibrate the raw confidence scores from the SVM classifiers to probabilities with the Platt [40] algorithm. Finally, we obtain linear OVA-SVM classifiers with compatible probability scores and use the parameters $\{w_{OVA}^{y_c}\}_{c=1}^C$ as initial input to the proposed method for structured output learning.

4.1.4. Evaluation metrics Given an unlabeled test image, we first compute the score for each label using the learnt model, and then select top-K scoring labels according to (4). For the

image classification task on the ILSVRC-10 dataset, we set $K = \{1, 2, 3, 4, 5\}$ and report the result of average *hierarchical error*, which measures the semantic distance between each predicted label and groundtruth label in the predefined WordNet hierarchy. Note that a small hierarchical error indicates better classification performance. For the image annotation task, we set $K = 5$ and use two standard criteria to evaluate the annotation performance: (i) average precision per label P, and (ii) average recall per label R. Note that the P and R scores are obtained by first computing precision and recall for each label and then averaging over all labels. In addition, as the number of labels in NUS-WIDE dataset is considerably small, we add another two criteria: *Hamming loss* and *Average AUC*, which take the performance of overall prediction and ranking into account. For the four evaluation metrics except *Hamming loss*, a larger numerical value indicates better annotation performance.

4.2. Evaluation on ILSVRC-10 dataset

For our first experiment, we consider structured classification on the ILSVRC-10 dataset to compare the proposed method with previous OVA-SSVM method. To calculate the image-specific structured loss of the proposed method, we need to handle a large number of images and feature vectors of high dimension to compute pairwise distance. It is intractable to compute the pairwise distance of all the training images and keep the data in onboard memory (experiment is run on an eight-core, 32 GB RAM desktop PC). According to the training scheme used in OVA-SSVM method, we randomly select 25% of the entire training data (half for training, half for validation); hence computing the image-specific structured loss on the subset of training data is feasible. We set the regularization parameter $\lambda = 10^{-4}$ when using SIFT features and $\lambda = 10^{-6}$ when using CNN features, and repeat the training scheme 10 times with different random selection of training data. As usual, we report the test error for the best performance on the validation set.

We compare our method with three baselines: OVA-SVM, OVA-SSVM (Flat) that uses flat structured loss, and OVA-SSVM (Hie) that uses hierarchical structured loss. Tables II and III show the test error of these methods with different number K of predicted labels using SIFT features and CNN features, respectively. It can be observed that (i) the proposed method and the two derivations of OVA-SSVM methods can reduce the test error of original OVA-SVM when a structured setting is adopted, showing the efficiency of structured output learning; (ii) the proposed method and OVA-SSVM (Hie) method are superior to OVA-SSVM (Flat) method, showing the importance of modeling the semantic relationships among output labels; (iii) the proposed method consistently obtains better performance on the more powerful CNN features compared with using SIFT features, which implies the proposed method is adaptive to different type of visual features; and (iv) the proposed method performs on par with OVA-SSVM (Hie) for different value of K. It indicates that the proposed image-specific structured loss function is an alternative to the hierarchical structured loss of OVA-SSVM (Hie) and can also incorporate the semantic hierarchy of labels. However, the calculation of proposed image-specific loss is a data-driven approach, which is more advantageous than the predefined hierarchical cost used in OVA-SSVM (Hie) method. Therefore, it would be more efficient to apply the proposed image-specific loss when there is no predefined semantic hierarchy of the dataset.

Table II. Test error of the proposed method and baselines with different values of K using 1024-dimensional SIFT features

	K = 1	K = 2	K = 3	K = 4	K = 5
OVA-SVM	11.35	9.29	8.08	7.25	6.64
OVA-SSVM (Flat)	11.24	9.21	8.00	7.24	6.58
OVA-SSVM (Hie)	10.88	8.85	7.71	6.93	6.36
Proposed	10.90	8.83	7.69	6.94	6.38

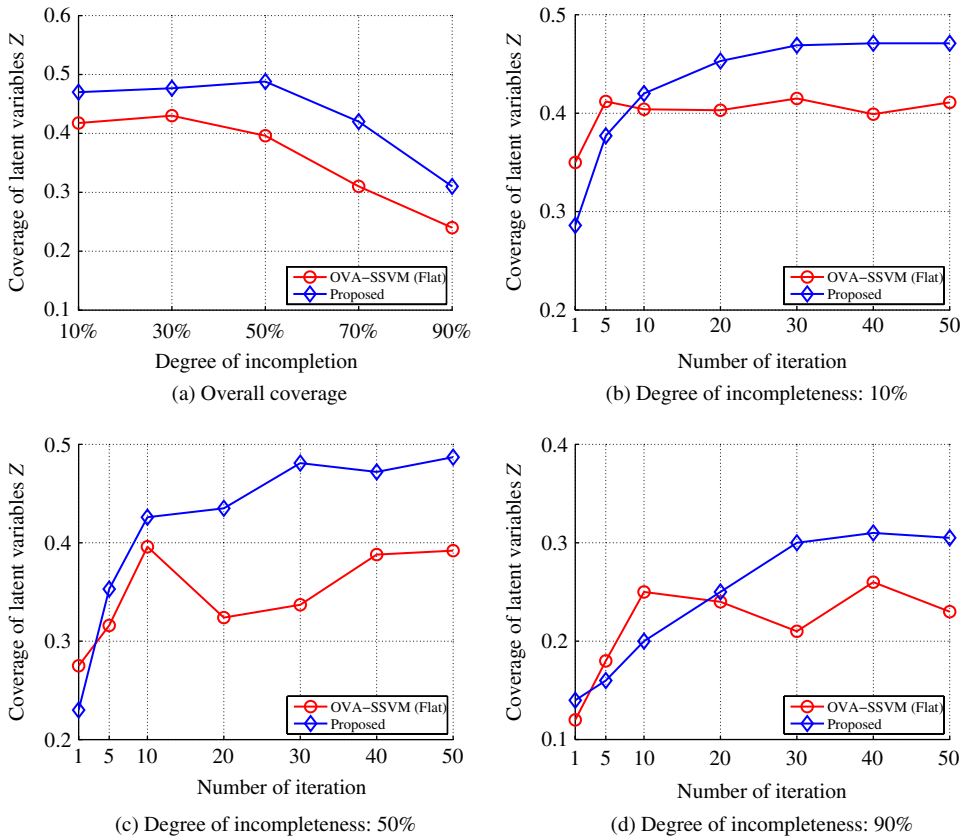
Table III. Test error of the proposed method and baselines with different values of K using 4096-dimensional CNN features

	K = 1	K = 2	K = 3	K = 4	K = 5
OVA-SVM	9.20	7.23	6.19	5.49	5.00
OVA-SSVM (Flat)	9.05	7.12	6.11	5.44	4.98
OVA-SSVM (Hie)	8.91	6.94	5.90	5.33	4.83
Proposed	8.85	6.91	5.88	5.32	4.85

Figure 5 gives samples of the classification results of the proposed method and three baselines, with the predicted labels of baselines (the third row) and the proposed method (the last row). From the samples we can see that, in most cases, the proposed method not only gives the correct prediction corresponding to the groundtruth (the second row) but also considers the structured relationships among the predicted labels to assign semantically related labels whose distances are smaller to the groundtruth compared with the baselines. For example, the predicted labels {"vestment", "gown", "academic gown"}, {"sloth bear", "brown

Groundtruth: vestment	Groundtruth: half track	Groundtruth: fire engine	Groundtruth: sloth bear	Groundtruth: planetarium	Groundtruth: shin guard
OVA-SVM: *vestment* (0), kimono (6), academic gown (3), thresher (11), guillotine (11)	OVA-SVM: Model T (5), thresher (10), steam locomotive (5), *half track* (0), covered wagon (6)	OVA-SSVM (Flat): *fire engine* (0), thresher (10), half track (5), Model T (4), toyshop (11)	OVA-SSVM (Flat): brown bear (3), apiary (16), lessa panda (7), mustard (18), miniature poodle (7)	OVA-SSVM (Hie): steam locomotive (11), garbage truck (11), dustcart (11) Model T (11), toyshop (7)	OVA-SSVM (Hie): pajama (6), *shin guard* (0), hair spray (11), stelis (16), buttress (11)
Proposed: *vestment* (0), gown (3), fancy dress (6), kimono (6), academic gown (3)	Proposed: amphibian (5), *half track* (0), plow (5), thresher (10), tank (2)	Proposed: *fire engine* (0), ambulance (4), police van (3), garbage truck (3), Model T (4)	Proposed: *sloth bear* (0), brown bear (3), American black bear (3), bearskin (16), Tibetan mastiff (7)	Proposed: *planetarium* (0), dome (11), greenhouse (5), mosque (5), tile roof (11)	Proposed: sock (6), knee pad (2), *skin guard* (0), Loafer (7), buttress (11)

Fig. 5. Samples of classification results on the ILSVRC-10 dataset. For the proposed method and three baselines, the top five scoring labels are selected as finally predicted labels. Note that the semantic distance ($d(y, y^n)$ in (8) between each predicted label with the groundtruth is shown in the brackets. And the correctly predicted labels that match the groundtruth are marked in italic font



(a) Overall coverage

(b) Degree of incompleteness: 10%

(c) Degree of incompleteness: 50%

(d) Degree of incompleteness: 90%

Fig. 6. Evaluation of coverage of latent variables with various degree of incompleteness. (a) Overall coverage, (b) Degree of incompleteness: 10%, (c) Degree of incompleteness: 50%, (d) Degree of incompleteness: 90%

bear", "American black bear"}, and {"sock", "knee pad", "skin guard"} reflect the semantic hierarchy and correlations of these labels.

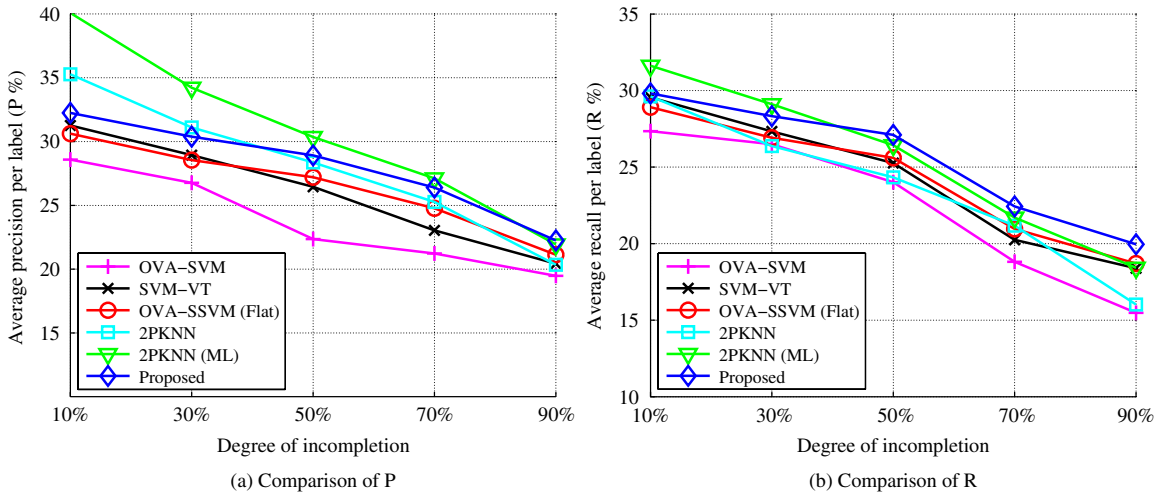
4.3. Evaluation on IAPRTC-12 dataset

4.3.1. Assessment of assigning latent variables. We now turn to evaluate the proposed method on image annotation task in this and the next section. Because of the more manageable size of the two datasets IAPRTC-12 and NUS-WIDE, we can directly compute the proposed image-specific structured loss for each training image in these two datasets. Since these two datasets have no predefined semantic hierarchy, the hierarchical structured loss is invalid for evaluation. We first consider the proposed method and OVA-SSVM (Flat) method to compare the efficiency of structured output learning with latent variables. Intuitively, the latent variables should capture those objects that appear in an image, but are not present in the groundtruth. Specifically, we explore how closely the assigned latent variable Z^n matches those labels $\Omega^n \backslash Y^n$ deleted from the originally full annotations of image x^n when following the learning algorithm depicted in Fig. 4. We use a measure termed $Coverage = \frac{1}{N}\sum_{n=1}^{N}\frac{|Z^n \cap (\Omega^n \backslash Y^n)|}{|Z^n|}$ to represent the averaged intersection between Z^n and $\Omega^n \backslash Y^n$ for all training image $x^n \in \mathcal{X}$. Note that a higher coverage indicates better assignments of latent variables.

Figure 6(a) shows the overall coverage of latent variable to the deleted labels in the full annotations with different degree of incompleteness. It can be observed that (i) the coverage of latent variable of both methods increases when the degree of incompleteness decreases, and this is reasonable because the more labels we have, the better we can predict the missing labels; (ii) our proposed method consistently obtains higher coverage for missing labels than OVA-SSVM (Flat) which simply uses flat structured loss, as the image-specific structured loss used in our method is more efficient in exploiting various contextual information of labels and images under the setting of incompleteness. Furthermore, in Fig. 6 (b)–(d), we explicitly demonstrate the changing of coverage of the latent variables through the iterations (as described in Fig. 4) under different degrees of incompleteness: 10%, 50%, 90%. We can learn that the proposed image-specific structured loss is appropriate to ensure our method to perform robustly, while OVA-SSVM (Flat) seems to be unstable through the iterations and results in inferior coverage. Especially, when the degree of incompleteness is pretty high (e.g. 70%, 90%), the coverage of the proposed method is significantly better than OVA-SSVM (Flat), showing the superiority of the proposed method under the setting of incompleteness.

4.3.2. Overall comparison with various degrees of incompleteness To make a comprehensive comparison, we first explore the labeling results from the binary classifiers: OVA-SVM and SVM-VT [24] (OVA-SVM combined with proposed image specific label relatedness as depicted in Section 3.1, without structured output learning), and then boost the binary classifiers by structured output learning via flat structured loss (OVA-SSVM (Flat)) and image-specific structured loss (the proposed method). In addition, we also take the method termed 2PKNN [14] into account, a nearest neighbor-based method that achieves the state-of-the-art performance for image annotation task with full labeling. We evaluate the basic 2PKNN method and its metric learning version 2PKNN (ML) under the same settings of incompleteness.

Figure 7 shows the annotation performance of these methods in terms of P and R with various degrees of incompleteness. Firstly, it can be seen that as the degree of incompleteness decreases, the performance of all the methods becomes better, since more labels for training data provides more information. Second, our method can boost the performance of binary classifiers



(a) Comparison of P (b) Comparison of R

Fig. 7. Comparison of annotation performance with various degree of incompleteness. (a) Comparison of P and (b) Comparison of R

OVA-SVM and SVM-VT under the setting of incompleteness, which verifies the efficiency of the incrementally structured output learning. Third, regarding the structured output learning stage, the proposed method performs remarkably better than OVA-SSVM (Flat), which uses the flat structured loss, especially when the degree of incompleteness is considerably high (e.g. 70%, 90%). The reason behind this is that we use more appropriate structured loss, which efficiently accounts for the dependencies between the predicted labels under the setting of incompleteness. Lastly, the 2PKNN method performs better than the proposed method when the degree of incompleteness is not high (e.g. 10%, 30%, 50%), as the elaborately designed neighbor selection scheme in 2PKNN method can efficiently construct a balanced neighborhood in this case, and the metric learning version can further capture the importance of different labels based on measured image similarity. However, when the degree of incompleteness is considerably high (e.g. 70%, 90%), the 2PKNN method performs on par with (even worse) than the proposed method. We believe that large proportions of missing labels from training images prevent the 2PKNN method from obtaining a balanced neighborhood and learning appropriate distance metric for similarity measure. Therefore, it indicates that the proposed method is more robust to tackle the issue of incompleteness.

4.4. Evaluation on NUS-WIDE dataset

Finally, to perform the experiment on the incompletely labeled NUS-WIDE dataset, besides the four methods compared above, we also consider state-of-the-art annotation methods that are designed for full labeling and incomplete labeling. The methods for full labeling include JEC [12], Tagprop [13], 2PKNN [14], and M3L [6]. In particular, for Tagprop and 2PKNN, we use their metric learning version, denoting them as Tagprop (ML) and 2PKNN (ML). The methods for incomplete labeling consist of SVM-VT [24], MLR-GL [23], Fasttag [25] and LEML [26]. To make a fair comparison, we use the code provided by the authors and follow the instructions in the corresponding papers to tune the model parameters.

Table IV shows the annotation performance of various methods depicted above. And we can make the following observations: (i) The proposed method consistently boosts the binary SVM classifiers (OVA-SVM and SVM-VT) and also obtain better performance than OVA-SSVM (Flat). (ii) The conventional nearest neighbor-based methods such as JEC, Tagprop, and 2PKNN fail to achieve comparable performance. This reason may be that the noisy and missing label assignments of NUS-WIDE dataset affects the results of selecting representative neighborhood and learning appropriate distance metric. (iii) The annotation methods including the proposed method designed for incomplete labeling are generally superior to conventional annotation methods with full labeling, which again addresses the significance of tackling the issue of incompleteness of practical annotation data. (iv) The proposed method performs comparably or better than even the recently proposed methods with incomplete labeling, which corroborates the efficiency of structured output learning on capturing the semantic correlations of labels when labels are incomplete. And it explores more efficient SVM-based method combined with structured output learning and provides another aspect for the image annotation problem with incomplete labeling.

Table IV. Annotation performance comparison among different methods on the NUS-WIDE dataset. Previous and our best results are highlighted in bold

Method	P (%)	R (%)	Average AUC	Hamming loss
JEC [12]	11.9	16.6	0.557	0.083
Tagprop (ML) [13]	13.2	23.8	0.707	0.074
2PKNN (ML) [14]	14.5	23.9	0.726	0.073
OVA-SVM	12.3	22.8	0.782	0.079
M3L [6]	16.1	23.2	0.791	0.071
SVM-VT [24]	16.7	24.3	0.806	0.069
MLR-GL [23]	14.2	23.5	0.722	0.078
Fasttag [25]	**18.4**	21.3	**0.834**	0.067
LEML [26]	17.5	24.6	0.798	0.076
OVA-SSVM (Flat) [30]	16.9	24.1	0.772	0.070
Proposed	17.7	**25.6**	0.819	**0.064**

Figure 8 gives qualitative samples of annotation results predicted by the proposed method on the two datasets. In particular, for the IAPRTC-12 dataset, we preserve the original training images without the deletion process to evaluate the generalization of the proposed method. From the samples, we can see that, although there are fewer groundtruth labels, our method can still make correct predictions on them. In addition, our method can also reflect semantic connectedness among the predicted labels, e.g. {"field", "landscape"}, {"restaurant", "building", "street"}, {"beach", "sand", "ocean"}, etc. This further demonstrates the effectiveness of the proposed method using structured output learning.

5. Conclusion and future work

In this paper, to tackle the issue of incomplete labeling, we leveraged the structured output learning method to boost the

(a) Field, mountain mountain, cloud, sky, field, landscape

(b) Front, people, street restaurant, building, street, people, front

(c) Stand fruit, people, woman stand, clothes

(d) Road llama, gravel, road stone, shrub

(e) Sand, sky beach, sand, sky, rocks, ocean

(f) Flowers, sun sky, tree, clouds, flowers, sun

(g) Cars vehicle, cars, nighttime, buildings, reflection

(h) Sand sky, sand, animal, person, horse

Fig. 8. Samples of annotation results of the proposed method on IAPRTC-12 ((a)–(d)) and NUS-WIDE ((e)–(h)). The red labels are the groundtruth, and the black ones are the top five labels predicted by the proposed method

performance of conventional OVA-SVM classifiers and formulated an image-specific structured loss function that is more appropriate to explore the dependencies of predicted multiple labels. We further developed an efficient optimization algorithm with lower computational complexity to learn the model parameters. Experimental evaluation demonstrated that the proposed image-specific structured loss could directly learn the semantic relationships of labels from the training data without a predefined semantic hierarchy. Furthermore, the proposed method is efficient to handle the issue of incomplete labeling and performs superior to several existing methods on the image annotation task. In the future, we will explore more efficient strategies of calculating the image-specific structured loss term for large quantities of training data, and we also plan to extend our method to the scenario where even some of the incomplete labels are incorrectly assigned to the training samples. This, in turn, would facilitate the annotation model to be robust against the defection of training data.

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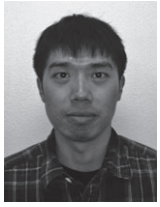

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